1D CNN — Two-Hand Dynamic Gesture Run Report

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Data paths

Training data root	/home/jestin/ThesisRepo/ML/NewTrainingData/Dynamic
Batch-test data root	/home/jestin/ThesisRepo/ML/NewTestData/Alan/Dynamic
Saved model path	saved_models/cnn_model_2026-05-05_09-10-13_fbca24f98e8b.keras
Experiment log	saved_models/experiment_results.csv

Sensor selection

Hands	left, right
Segments	wrist, palm, thumb, index, middle, ring, pinky
Features	YPR, Accel, Flex
Channels	176 (sequence length 90)

Preprocessing & split

Resample steps	90
Butterworth	on · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	minmax
Test size / seed	0.1 · random_state=42 · stratify=True
Sample counts	total=420 · train pool=378 · train+aug=2268 · hold-out=42
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll

Augmentation (training only)

Enabled	yes
Copies per sample	5
Random seed	42
Active augmentations	amplitude_scale, baseline_offset
• amplitude_scale	apply_prob=0.5, scale_range=(0.95, 1.05)
• baseline_offset	apply_prob=0.5, offset_std_ratio=0.02

Model architecture & training

Architecture	Conv1D(32, k=4, ReLU) → MaxPool(2) → Conv1D(64, k=4, ReLU) → MaxPool(2) → Flatten → Dense(64, ReLU) → Dropout(0.3) → Dense(softmax)
Optimizer / loss	adam · sparse_categorical_crossentropy
Input shape	(90, 176)
Trainable params	113,255
Epochs · batch size · val split	15 · 16 · 0.2

Cross-validation

Folds · shuffle	5 · True
Mean ± std val acc	0.9974 ± 0.0053
Best fold	1

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1812	76	1.0000	1.0000	1.0000
2	1812	76	1.0000	1.0000	1.0000
3	1812	76	0.9868	0.9868	0.9868
4	1818	75	1.0000	1.0000	1.0000
5	1818	75	1.0000	1.0000	1.0000

Final training & hold-out test

Final train acc / loss	0.9989 · 0.0027
Final val acc / loss	0.9912 · 0.0095
Best val acc / lowest val loss	0.9912 · 0.0095
Hold-out test acc / loss	1.0000 · 0.0000

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	6
Double_Nothing	1.000	1.000	1.000	6
Double_Pistol_Recoil	1.000	1.000	1.000	6
Double_Raise	1.000	1.000	1.000	6
Double_Wiggle	1.000	1.000	1.000	6
Double_cmere	1.000	1.000	1.000	6
Drum_Roll	1.000	1.000	1.000	6
macro avg	*1.000*	*1.000*	*1.000*	*42*

weighted avg	*1.000*	*1.000*	*1.000*	*42*

Batch test (NewTestData)

Class	Correct	Total	Accuracy
Double_Lower	0	5	0.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	**22**	**35**	**0.629**